Final Report

The Art of Scientific Computing: Complex Systems

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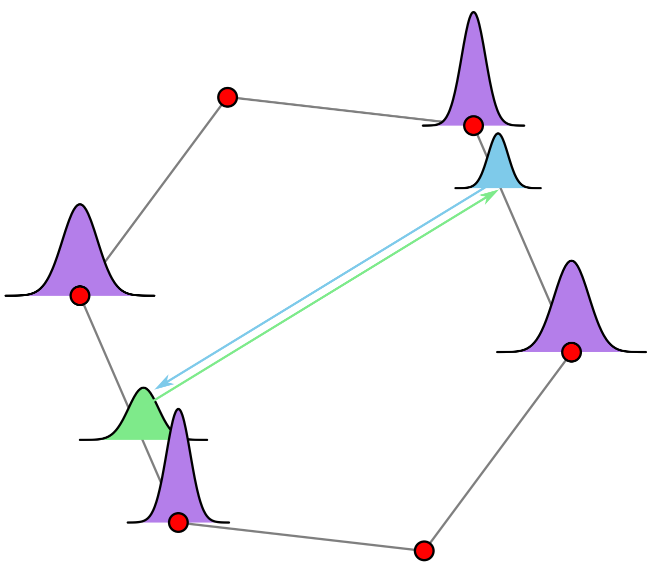

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Chapter 1

Introduction

This chapter introduces the background of self-organized criticality (SOC) and the Bak–Tang–Wiesenfeld (BTW) sandpile model. It also explains the theoretical importance of the Abelian property and the main aim of this project.

1.1 Complex Systems and Self-Organized Criticality

In many physical systems, the size of an effect is often related to the size of its cause. For example, there may only be a local impact when a small meteorite hits the Earth, but a much larger object may influence the whole orbital system. However, in complex system, this simple relationship does not always hold. In some complex systems, a small local perturbation can lead to a large-scale event [4].

A complex system usually contains many components which could interact with each other. Since these components affect each other, the global behaviour of the system is difficult to predict only from the local rules [4]. And some systems can produce events with very different sizes. For example, a small change of stress may cause a large earthquake; a small ignition source may cause a large forest fire; a small fluctuation in a financial market may also spread and lead to a wider market crash.

Bak, Tang, and Wiesenfeld introduced the idea of self-organized criticality (SOC) in 1987 [2]. The SOC describes systems which can automatically evolve to a critical state, without tuning external parameters. In the critical state, the system can produce both small and large events. The system also often shows scale-invariant behaviour [7].

Scale invariance means that, at different scales, the statistical behaviour of the system is similar [7]. And scale invariance is closely related to power-law behaviour [1]. This means that small events happen very frequently, while large events happen less frequently but they are still possible. For example, this type of behaviour is commonly observed in earthquakes, forest fires, and financial market fluctuations [1] [7].

Mathematically, a power-law distribution can arise from scale-type variable transformations [6]. If the probability measure is preserved under a change of variable,

$$P_X(x)\,dx = P_Y(y)\,dy,$$

then the transformed distribution depends on the Jacobian of the transformation. For a scale-type transformation such as

$$y = x^{-1/\alpha}, \qquad \alpha > 0,$$

the transformed probability density becomes

$$P_Y(y) = P_X(x)\left|\frac{dx}{dy}\right|.$$

If $P_X(x)$ is approximately constant near the origin, this gives

$$P_Y(y) \propto y^{-(1+\alpha)},$$

which is a power-law distribution.

This shows that scale-related transformations can naturally produce power-law behaviour. Therefore, power-law distributions are a natural description for systems with scale-invariant behaviour. (**Ans. to Q1**)

For studying SOC, the Bak–Tang–Wiesenfeld (BTW) sandpile model is a simple model to begin with. Although its rules are simple, the sandpile model can produce complex avalanche behaviour. By adding grains of sand slowly and applying local toppling rules, the model can reach a critical state and show statistics with power-law [2]. Therefore, the BTW model is useful for us to study complex systems, self-organized criticality, and non-equilibrium statistical behaviour.

1.2 The Bak–Tang–Wiesenfeld (BTW) sandpile model

The Bak–Tang–Wiesenfeld (BTW) sandpile model simulates a real sandpile on a finite two-dimensional lattice. Although the real sandpile is not created exactly on a the grid, this simplified model keeps the main mechanisms of the system, which include slow input, local instability, chain reactions, and boundary loss [2].

In the BTW model, the sandpile is performed on an $N \times M$ grid. Each site (i, j) has a height $h(i, j)$ as an integer, which represents the number of sand grains at that site. At each time step, one grain of sand is added to one site. This process represents a slow and local external driving force [8].

When the height of a site reaches or exceeds the given stability threshold, the site becomes unstable. In the standard two-dimensional BTW model, the threshold is usually denoted as 4. If a site satisfies:

$$h(i, j) \geq 4,$$

then this site topples.

During a toppling process, the site loses 4 grains of sand, and these grains are assigned to its four orthogonal neighbours: up, down, left, and right [2]. This rule can be mathematically written as

$$h(i, j) \leftarrow h(i, j) - 4,$$

$$h(i \pm 1, j) \leftarrow h(i \pm 1, j) + 1,$$
$$h(i, j \pm 1) \leftarrow h(i, j \pm 1) + 1.$$

If a toppling occurs at the grid boundary, some grains may be sent outside the lattice, and these grains will be removed from the system and counted as boundary loss. The boundary loss is important, because it prevents the total amount of sand on the grid from increasing without limit. With continuous sand addition and boundary loss, the system can eventually reach a statistically stable state [2] [8].

The toppling of one site may cause an unstable condition for its neighbours, and these kind of sites may topple and create further unstable sites. Then a chain reaction of topplings happens, and this event caused by adding one grain of sand is called an avalanche [8]. The behaviour of avalanches is the main object studied in this project, because is shows how a small local input can create events with different sizes.

According to above introduction for the BYW model, we can observe several important properties required for a system to exhibit SOC: First, the BTW model is a slowly driven system, we only add one grain of sand to the lattice at each time step; Second, the model contains a local threshold mechanism, since the site topples when the height of a site reaches 4 or more; Third, the BTW model can naturally produces avalanche behaviour, since one toppling event may cause neighbouring sites to become unstable and then create a chain reaction of more topplings. Fourth, the model includes boundary dissipation, since some grains leave the system when toppling occurs at the edge of the lattice; Fifth, the BTW model can evolve toward a critical state without carefully tuning parameters [2]; Finally, the BTW model exhibits scale invariance and power-law behaviour [2]. (**Ans. to Q2**)

In the BTW model, finite boundaries are important because they allow sand grains to leave the system when toppling occurs. This boundary dissipation can prevent the total amount of sand from increasing without limit. As a result, the system can reach a statistically stable critical state under continuous sand addition. However, if we consider infinite lattice and drop sand on only a finite region, grains could spread arbitrarily far away. (**Ans. to Q3**)

In the simulation in this study, four avalanche properties are recorded: topples, area, loss, and length. The value of topples is the total number of toppling events in one avalanche; The value of area is the number of unique sites that topple at least once. The value of loss is the number of grains that leave the lattice through the boundary. The value of length measures the spatial extent of the avalanche, in this study, length is defined as the avalanche radius.

By implementing above sand addition, toppling, and avalanche simulation, we can study the behaviour of the system, the distributions of avalanche properties, the correlations between variables.

1.3 Abelian Property and Theoretical Importance

Besides studying avalanche through simulation, the BTW sandpile model also has an important theoretical property, which is called the Abelian property. This property means that, during one avalanche, the order of topplings does not influence the final relaxed configuration. If there are several unstable sites at the same time, these sites can be toppled in different orders. Dhar states that toppling site i before site i' gives the same configuration as toppling site i' before site i, so the toppling operations commute [3]. Therefore, when all necessary topplings are completed, the final stable sandpile configuration does not depend on the chosen toppling order.

This property is important for the simulation. In one avalanche, there may be many topplings, and there may be more than one unstable site at the same time. If the model did not have the Abelian property, the code would need to control the exact order of topplings very carefully. This would make the simulation more difficult to implement. However, because the BTW model has the Abelian property, the avalanche can be processed using a simpler algorithm.

The Abelian property is also important from a theoretical aspect. It shows that the BTW model has a clear mathematical structure, even though its local rules are simple. In the project document, this property is introduced by using the toppling matrix, the toppling operator, and the stabilization operator. The toppling matrix describes how the height variables are updated when a site topples [3]. In our project notation, the toppling operator represents this local update from one configuration to another. The stabilization operator then represents the full relaxation process, where an unstable configuration is turned into a stable configuration through a sequence of topplings [3].

Therefore, the abelian property directly supports the code design and the reliability of the simulation results in this project. This allows the simulation to record avalanche properties such as topples, area, loss, and length in a consistent way.

1.4 Aim of The Project

The main aim of this project is to implement a Bak–Tang–Wiesenfeld (BTW) sandpile model, and use it to study self-organized criticality and the related statistics on a finite two-dimensional lattice. At the end, we will extend the model to investigate different system behaviours.

Chapter 2

Model Description

In this project, we use the BTW Sandpile Model to study self-organized criticality (SOC). The model uses a finite two-dimensional lattice to perform a sandpile system, and each lattice site corresponds to a local sand pile. In this chapter, we will introduce a detailed mathematical definition for the sandpile model, including the local rules. We will also discuss the theoretical questions about the Abelian property.

2.1 The BTW Sandpile Model

The two-dimensional lattice is defined as:

$$L = \{1, \ldots, N\} \times \{1, \ldots, M\}$$

where N is the number of rows, M is the number of columns, $(i, j) \in L$ represents a lattice site.

At time step t, the number of sand grains at site (i, j) is given by:

$$p_t(i, j) \in \mathbb{N}_0.$$

The full sandpile configuration at time t is written as:

$$P_t = \{p_t(i, j)\}_{(i,j) \in L}.$$

In this model, t represents the external sand addition step, rather than the number of internal topplings during an avalanche. At each time step, one grain of sand is added to the system, while the internal toppling process is considered as instantaneous [5].

2.2 Toppling Dynamics and Avalanche Process

At each time step t, we randomly select a lattice site $(i_t, j_t) \in L$, and then add one grain of sand on it:

$$p_t(i_t, j_t) \leftarrow p_t(i_t, j_t) + 1.$$

The model uses a fixed stability threshold:

$$k = 4.$$

A site becomes unstable when:

$$p_t(i, j) \geq 4.$$

When a site becomes unstable, a toppling occurs on it. During one toppling process, the site loses four grains of sand, and one grain is transferred to each of its four nearest neighbours [5]:

$$p_t(i, j) \leftarrow p_t(i, j) - 4,$$

$$p_t(i \pm 1, j) \leftarrow p_t(i \pm 1, j) + 1,$$
$$p_t(i, j \pm 1) \leftarrow p_t(i, j \pm 1) + 1.$$

Note that we only consider the nearest neighbours, without considering diagonal neighbours [5]. The nearest neighbour set is defined as:

$$\mathcal{N}(i,j) = \{(i+1,j), (i-1,j), (i,j+1), (i,j-1)\}$$

In this project, we use the open boundary condition. If a toppling happens at the boundary of the lattice, some grains may be removed outside the lattice and system.

A single grain addition may cause a chain reaction of topplings. When one unstable site topples, its neighbouring sites may also become unstable and then begin toppling. This process is called an avalanche. In this project, an avalanche is defined as the full sequence of topplings caused by one grain addition until the system becomes stable [5], which is given by:

$$p_t(i,j) < k, \quad \forall (i,j) \in L.$$

2.3 Avalanche Properties

In order to study the avalanche behaviour, this project considers four avalanche properties: topples, area, loss, and radius [5].

We denote the avalanche topples as s, which is the total number of topplings during one avalanche. Note that if one site topples multiple times during the same avalanche, then every toppling is counted in s; We denote the avalanche area as a, which is the number of unique lattice sites that topple during one avalanche; We denote the avalanche loss as l, which is the total number of sand grains that leave the lattice during one avalanche.

We denote the avalanche length as r, which is defined as the spatial radius:

$$r = \max_{x \in A} \|x - x_0\|_2$$

where x_0 is the dropping site of the avalanche, A is the set of all toppled lattice sites, $\|\cdot\|_2$ represents the Euclidean distance.

This length measures the maximum Euclidean distance between the dropping site and all toppled sites, which represents how far the avalanche spreads on grid. And we use the Euclidean distance because it gives a simple geometric measure of spatial spreading and treats diagonal spreading naturally. Therefore, this definition can provide a suitable measure of avalanche length.

After each avalanche, the simulation records the corresponding values of avalanche properties as (s, a, l, r), which is used for the following statistical analysis.

2.4 Example of an Avalanche Process

In order to clearly illustrate the toppling dynamics and demonstrate how to calculate avalanche properties in the BTW sandpile model, we use a 3×3 lattice avalanche example:

$$\begin{bmatrix} 1 & 3 & 3 \\ 2 & 4 & 2 \\ 2 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 & 3 \\ 3 & 0 & 3 \\ 2 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & 4 \\ 3 & 1 & 3 \\ 2 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 0 \\ 3 & 1 & 4 \\ 2 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}$$

Here we assume the dropping site is $(2, 2)$. The avalanche process of this example is performed by implementing the definition and rules as shown above. Then we calculate the avalanche properties by following the definition.

First, the avalanche topples s denote the total number of toppling events during the avalanche. In this example, four topplings happen at the lattice sites: $(2,2)$, $(1,2)$, $(1,3)$, $(2,3)$, then

$$s = 4.$$

Next, the avalanche area a denotes the number of unique lattice sites that topple during the avalanche. Since all four toppled sites are different:

$$a = 4.$$

The avalanche loss l denotes the number of sand grains that leave the system through the lattice boundary. In this example, $(2,2)$ is an interior site and produces no loss, $(1,2)$ and $(2,3)$ produce one grain loss respectively, $(1,3)$ produce two grain losses. Then, the total loss is:

$$l = 4$$

Finally, the avalanche radius r is defined as the maximum Euclidean distance between the dropping site and all toppled sites. Since the dropping site is $x_0 = (2,2)$ and toppled sites are $(2,2)$, $(1,2)$, $(1,3)$, $(2,3)$, the corresponding Euclidean distances are $0, 1, 1, \sqrt{2}$. Then

$$r = \sqrt{2}$$

Hence, the avalanche observables for this example are:

$$s = 4, \quad a = 4, \quad l = 4, \quad r = \sqrt{2}.$$

2.5 Abelian Property

This section further discusses the theoretical questions related to the Abelian property. In particular, we introduce the toppling matrix, the toppling operator, and the stabilization operator, and explain how to use them to describe the Abelian property in the BTW model.

To describe the avalanche relaxation process, we introduce the following mathematical definitions. For each site $x \in L$, the sand height is represented by the function

$$p : L \to \mathbb{N}^0.$$

A configuration is called stable if $p(x) < k$ for all $x \in L$, where k is the stability threshold. If there exists at least one site satisfying $p(x) \geq k$, then the configuration is unstable. The toppling matrix $\Delta(x, y)$ denotes the redistribution of sand from site x to site y during a toppling event. The toppling matrix must satisfy the following conditions:

$$\begin{aligned}
\forall x, y \in L \text{ where } x \neq y, \quad & \Delta(x, y) = \Delta(y, x) \geq 0, && \text{(a)}\\
\forall x \in L, \quad & \Delta(x, x) < 0, && \text{(b)}\\
\forall x \in L, \quad & \sum_{y \in L} \Delta(x, y) \leq 0, && \text{(c)}\\
& \sum_{x, y \in L} \Delta(x, y) < 0. && \text{(d)}
\end{aligned}$$

Condition (a) means that the sand transfer between different sites is symmetric and non-negative. This guarantees consistent interactions between neighbouring sites; Condition (b) means that the toppled site loses sand during relaxation. Without this condition, the configuration could not stabilize; Condition (c) states that one toppling event does not increase the total amount of sand in the system. Sand is either conserved or lost through the boundary; Condition (d) guarantees that the system has net dissipation. For a finite lattice with open boundaries, sand can leave the system during boundary topplings, which prevents unlimited accumulation of sand. (**Ans. to Q9**)

In the BTW model, the toppling matrix is defined as:

$$\Delta(x, y) = \begin{cases} -2d, & x = y, \\ 1, & x, y \text{ are nearest neighbours}, \\ 0, & \text{otherwise}. \end{cases}$$

Here, d is the dimension of the lattice. In a d-dimensional square lattice, an interior site has $2d$ nearest neighbours. Then, when site x topples, it loses $2d$ grains of sand, each nearest neighbour receives one grain, and all other sites are not affected. According to the definition form, the stability threshold is $k = 2d$.

We now verify that this definition satisfies the four conditions. For condition (a), if x and y are nearest neighbours, then

$$\Delta(x, y) = \Delta(y, x) = 1.$$

If they are not neighbours, then

$$\Delta(x, y) = \Delta(y, x) = 0.$$

Therefore, for $x \neq y$,

$$\Delta(x, y) = \Delta(y, x) \geq 0.$$

For condition (b), by definition,

$$\Delta(x, x) = -2d < 0.$$

For condition (c), if x is an interior site, it has $2d$ neighbours, so

$$\sum_{y \in L} \Delta(x, y) = -2d + 2d \cdot 1 = 0.$$

If x is a boundary site, it has fewer than $2d$ neighbours inside the lattice. Let α_x be the number of its valid neighbours in L, and $\alpha_x \leq 2d$. Then

$$\sum_{y \in L} \Delta(x, y) = -2d + \alpha_x \leq 0.$$

For condition (d), since the lattice is finite and has open boundaries, there is at least one boundary site. For a boundary site x, we have $\alpha_x < 2d$, so

$$\sum_{y \in L} \Delta(x, y) = -2d + \alpha_x < 0.$$

Therefore, when summing over all sites, there is at least one negative contribution. Hence,

$$\sum_{x, y \in L} \Delta(x, y) < 0.$$

Thus, the BTW toppling matrix satisfies all four conditions. (**Ans. to Q10**)

Given the definition of the toppling matrix, the toppling operator T_x maps a configuration p to a new configuration p' is:

$$(T_x p)(y) := \begin{cases} p(y) + \Delta(x, y), & p(x) \geq k, \\ p(y), & \text{otherwise}. \end{cases}$$

Assume that both x and x' are unstable in the configuration p, i.e.,

$$p(x) \geq k, \qquad p(x') \geq k.$$

Since $\Delta(x,x') \geq 0$ for $x \neq x'$, toppling site x cannot decrease the height at site x'. Therefore, x' remains unstable after toppling x. Similarly, x remains unstable after toppling x'. Hence, both toppling orders are valid.

For any site $y \in L$,

$$(T_{x'}T_x p)(y) = p(y) + \Delta(x,y) + \Delta(x',y),$$

and,

$$(T_x T_{x'} p)(y) = p(y) + \Delta(x',y) + \Delta(x,y).$$

Since addition is commutative,

$$\Delta(x,y) + \Delta(x',y) = \Delta(x',y) + \Delta(x,y).$$

Then,

$$(T_{x'}T_x p)(y) = (T_x T_{x'} p)(y), \quad \forall y \in L$$

Hence,

$$T_{x'}T_x p = T_x T_{x'} p.$$

Thus, the toppling operators commute for unstable configurations. (**Ans. to Q11**)

In general, an unstable configuration may require more than one toppling before it becomes stable. Therefore, we define a stabilization operator to denote the whole relaxation process. This operator maps any height configuration to a stable height configuration,

$$\mathbb{T} : U_L \to S_L,$$

where U_L is the space of all possible height configurations, and $S_L \subset U_L$ is the subset of stable height configurations.

If an avalanche contains N topplings, then the stabilization operator can be written as,

$$\mathbb{T} = \prod_{i=1}^{N} T_{x_i},$$

where N is the number of instabilities in an avalanche.

We now show that the stabilization operator $\mathbb{T}$ is well defined. This means that, for a given unstable configuration p, the final stable configuration does not depend on the order of legal topplings.

Assume that two toppling sequences both stabilize p. Let n_x and m_x be the number of times site x is toppled in the first and second sequence, respectively. Then the two final configurations are

$$p'(y) = p(y) + \sum_{x \in L} n_x \Delta(x,y), \quad p''(y) = p(y) + \sum_{x \in L} m_x \Delta(x,y).$$

It is enough to show that

$$n_x = m_x, \quad \forall x \in L.$$

For contradiction, suppose that the second sequence topples some sites more times than the first sequence. Let v be the first site among them. Before this extra toppling at v, we have

$$m_v = n_v, \quad m_x \leq n_x, \quad \forall x \neq v.$$

Let ξ be the current configuration at this moment. Since the toppling at v is legal,

$$\xi(v) \geq k.$$

Now compare $\xi(v)$ with the stable configuration $p'(v)$:

$$\xi(v) - p'(v) = \sum_{x \in L} (m_x - n_x)\Delta(x,v).$$

Since $m_v = n_v$, the term with $x = v$ is zero. For $x \neq v$, we have $m_x - n_x \leq 0$ and $\Delta(x, v) \geq 0$. Then,

$$\xi(v) - p'(v) \leq 0,$$

so

$$\xi(v) \leq p'(v).$$

However, p' is stable, then

$$p'(v) < k.$$

Therefore,

$$\xi(v) < k,$$

which contradicts the fact that the toppling at v is legal. Then,

$$m_x \leq n_x, \quad \forall x \in L.$$

By exchanging the two sequences, we also obtain

$$n_x \leq m_x, \quad \forall x \in L.$$

Then,

$$n_x = m_x, \quad \forall x \in L.$$

Therefore,

$$p'(y) = p''(y), \quad \forall y \in L.$$

So the final stable configuration is unique. Hence, the stabilization operator $\mathbb{T}$ is well defined. (**Ans. to Q12**)

In summary, the Abelian property shows that the avalanche relaxation process in the BTW model is mathematically consistent. Therefore, in the simulation, it is not necessary to choose a specific order for processing unstable sites. As long as all legal topplings are completed, the final result is the same. This property can simplify the algorithm and also support the following analysis for the avalanche properties.

Chapter 3

Model implementation

This chapter introduces the numerical implementation of the BTW sandpile model. Then we explain how the program is used to simulate the evolution of the sandpile system and avalanche behavior. Then, we deliver the analysis for the state of systems and avalanche properties.

3.1 Code Structure

This project uses Python to implement the BTW sandpile model, and the program is designed according to the mathematical definitions in previous chapter. The code structure mainly consists of three parts: the core model class, the simulation control module, and the statistical analysis module.

First, we use the *SandPile* class to implement the main rules of the BTW model, including sand addition, toppling dynamics, boundary dissipation, and avalanche process. The function *drop_sand()* is used to add sand grains onto the lattice; the function *topple()* performs the toppling process for the unstable site; the function *avalanche()* repeatedly processes unstable sites until the whole system becomes stable; and the function *avalanche_length()* computes the avalanche radius. Further more, function *mass()* and function *is_stable()* are used to monitor the system state.

Second, we use the function *run_sandpile()* to perform the complete simulation procedure. This function controls the initialization for the lattice, setting and the random seed, and performs the burn-in process, then deliver the sampling simulation. During the burn-in process, the it adds sand grains and performs avalanche relaxation without recording statistical observables. During the sampling process, it records avalanche observables for the following statistical analysis.

Finally, the program contains functions for statistical analysis. The function *plot_average_height()* is used to analyze the evolution of the average height and study the statistically stationary state. The function *plot_powerlaw_distribution()* is used to fit the power-law distributions. The function *plot_gamma_distribution()* is used to fit the gamma distributions. The function *plot_correlation()* is used to analyze the statistical correlations between different avalanche properties.

3.2 Steady-State Behaviour

In order to study whether the BTW sandpile model reaches a stable state, we investigate the evolution of the average height of the system. The average height [5] is defined as:

$$\langle p_t \rangle = \frac{1}{NM} \sum_{i,j} p_t(i,j),$$

where $p_t(i,j)$ represents the number of sand grains at lattice site (i,j) at time step t.

Before simulating, we first give a simple guess about the behavior of $\langle p_t \rangle$. Since the lattice is initially empty, then very few sands leave the system through the boundary at the beginning. As sand grains are continuously added onto the lattice, the total mass of the system increases, then the average height is expected to increase at the early stage. However, as sand grains accumulate on the lattice, more sites reach the instability threshold and produce the avalanche. Then the boundary loss gradually increases, and the sand grains added into the system begin to balance with the sand grains leaving through the boundary. Therefore, we expect that the average height will not increase without limit, but approach a stable level and fluctuate around it.

To verify this behaviour, we use a two-dimensional lattice with $N = 64$, and perform 100000 sand addition steps. During this simulation, the system keeps performing grain addition and avalanche relaxation, while recording the average height at each time step. Figure 3.1 shows the evolution of the average height over time. At the beginning of this simulation, the average height increases rapidly. As the simulation continues, the average height reaches a level around 2.1, and then fluctuates around it instead of increasing without limit. This result shows that the system can gradually reach a steady state. In this state, the system still changes because avalanche processes and boundary losses continue to occur, but overall it remains relatively stable. (**Ans. to Q4**)

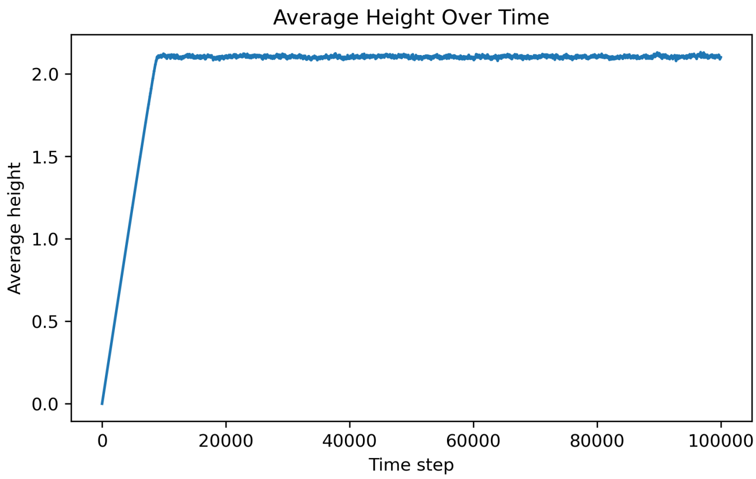


Figure 3.1: Average height over time

Although the average height becomes stable after a sufficiently long time, we should distinguish different types of stability. In this project, it is important to distinguish between a steady state, a statistically stationary state, and an equilibrium state: The steady state means that some macroscopic observables remain approximately constant over time; A statistically stationary state means that, although the microscopic configuration of the system continues changing, the statistical properties of the system remain stable over time; An equilibrium state is different from the two states above. In an equilibrium system, there is usually no continuous external driving force or dissipation process.

In this project, the statistically stationary state is mainly characterized by the behaviour of system observables. For example, the $\langle p_t \rangle$ fluctuates around a stable level after the initial growth stage, without showing long-term divergence. This behaviour indicates that the system could reach a statistically stationary state. (**Ans. to Q5**)

3.3 Avalanche Statistics and Correlations

Before studying avalanche statistics, it is necessary to choose a reasonable the sampling method. For complex systems, sampling a system over time and sampling an ensemble of systems are not always equivalent. Sampling a system over time means evolving the same system for a long time and continuously recording the system observables. In contrast, sampling an ensemble of systems means considering many independent systems with different initial conditions, and then averaging their statistical quantities. If the system behaviour depends on the initial condition for a long time, time sampling and ensemble sampling may give different statistical results.

To discuss whether these two sampling methods are equivalent, we need to introduce ergodicity. If a system is ergodic, then after sufficiently long time evolution, its time average will approach the ensemble average:

$$\text{time average} \approx \text{ensemble average}.$$

For the BTW sandpile model, we consider that the system is approximately ergodic. First, the system continuously performs grain addition and boundary dissipation, so the lattice configuration keeps changing. Second, after the burn-in stage, the dependence on the initial condition becomes much weaker, and the average height and avalanche activity gradually enter a statistically stationary state. (**Ans. to Q6**)

Therefore, this project uses time sampling to analyze avalanche statistics. In particular, after the burn-in process, the program continues running a single BTW sandpile system and records avalanche observables for the following statistical analysis. Then we study the statistical distributions of avalanche observables. The simulation uses a lattice with $N = 64$, and records 200000 avalanches after the burn-in process. The recorded observables are (s, a, l, r).

Figure 3.2 show the distributions of avalanche topples s and area a. The results show that these two variables are approximately linear on log-log plots, then we can fit them by using the power-law distribution, which is given by:

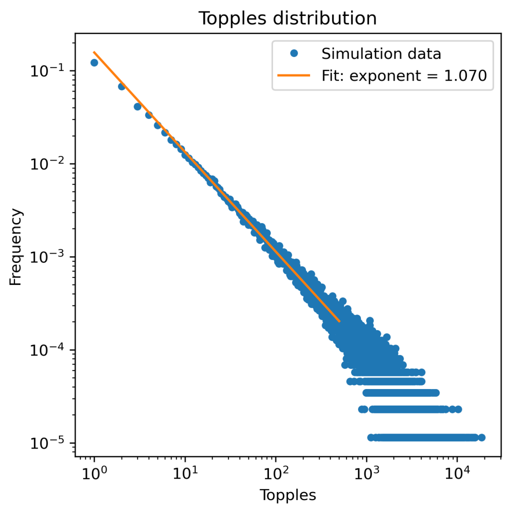
$$P(x) \propto x^{-\alpha},$$

where α is the power-law exponent. According to the fitting results, the exponent for topples s is approximately:

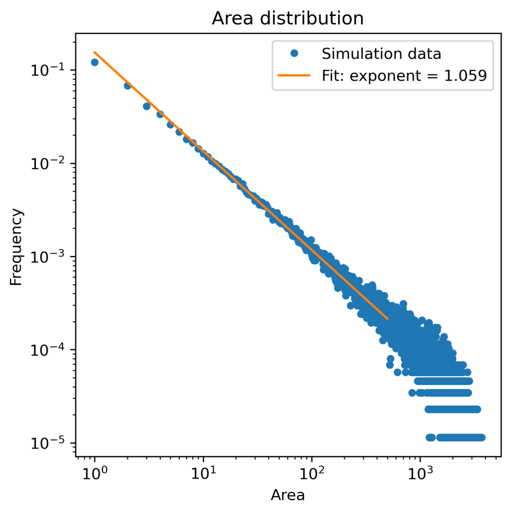
$$\alpha_s = 1.070,$$

and the exponent for area a is approximately:

$$\alpha_a = 1.059.$$



(a) Topples distribution



(b) Area distribution

Figure 3.2: Distribution fit for topples and area

In contrast, the distributions of loss l and radius r do not show very clear power-law behaviour. Then we consider using the Gamma distribution to fit them, as shown in Figure 3.3. The probability density function of the Gamma distribution can be written as:

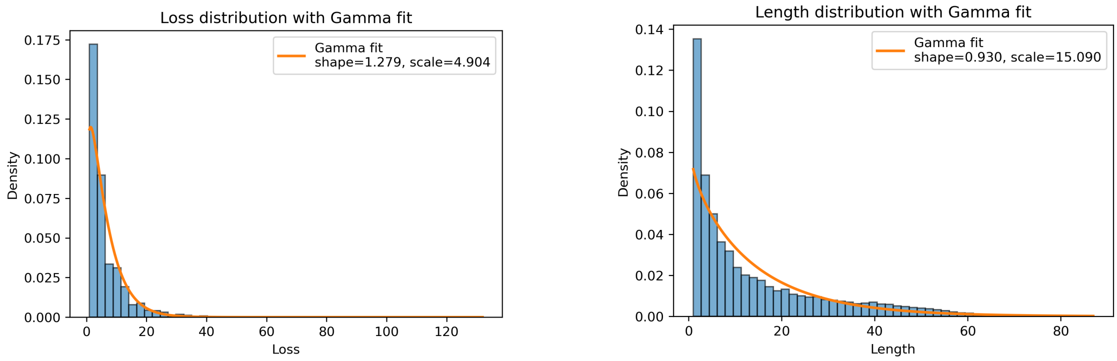
$$f(x; k, \theta) = \frac{x^{k-1}e^{-x/\theta}}{\Gamma(k)\theta^k}, \qquad x > 0,$$

where k is the shape parameter and θ is the scale parameter. According to the fitting results, the parameters for loss l are:

$$k_l = 1.279, \qquad \theta_l = 4.904,$$

and the parameters for radius r are:

$$k_r = 0.930, \qquad \theta_r = 15.090. \quad (\textbf{Ans. to Q7})$$



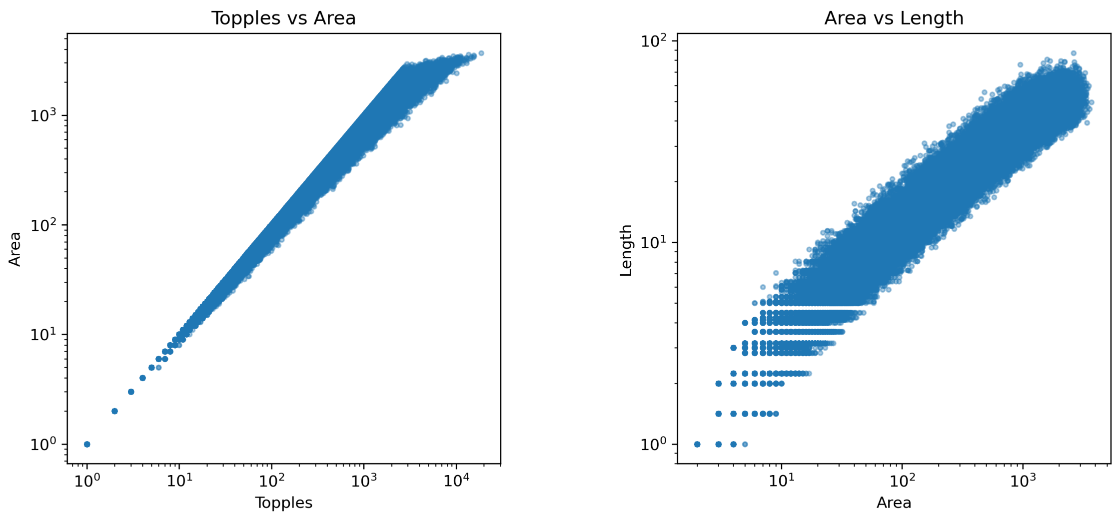
(a) Loss distribution

(b) Length distribution

Figure 3.3: Distribution fit for topples and area

Between the avalanche properties, We initially expect that s, a, and r should show clear positive correlations. The reason is that a larger avalanche contains more toppling events, then it is more likely to affect more lattice sites, which can produce a larger avalanche area. At the same time, as the avalanche spreads farther across the lattice, the avalanche radius also becomes larger.

The simulation results are shown in Figure 3.4, and they show agreement with the initial expectation — the correlation coefficients for topples & area and area & length are 0.926 and 0.890, respectively. Furthermore, topples & loss and topples & length also show relatively strong positive correlations which are 0.736 and 0.733 respectively. It indicates that avalanches with more toppling events are more likely to reach the lattice boundary and to spread far away with the dropping site, then produce larger boundary loss and length. (**Ans. to Q8**)



(a) Topples vs Area

(b) Area vs Length

Figure 3.4: Properties with high positive correlation

Chapter 4

Extension

To further study the effect of the driving mechanism on the BTW sandpile model, we consider a simple extension of the original model. In the original BTW model, only one sand grain is added to the lattice at each time step. In this extension, we add three sand grains to the same random lattice site at each time step, and the avalanche relaxation is performed after all grains have been added.

Before analyzing the simulation results, we first make some predictions for this extension. For the average height, since three sand grains are added at each time step, we expect sand accumulation to be faster at the early stage. Therefore, $\langle p_t \rangle$ is expected to increase faster than in the original model. However, the average height should not increase without limit. After a sufficiently long time, we still expect the system to enter a steady state.

From the simulation (as shown in Figure 4.1), the extension model is generally consistent with the previous prediction. At the early stage, sand accumulation in the extension model is clearly faster. Second, the average height of both models does not increase without limit. The final average height of the original model is approximately 2.1009, while the final average height of the extension model is approximately 2.1003. Overall, the extension model mainly changes the speed of reaching the steady state.

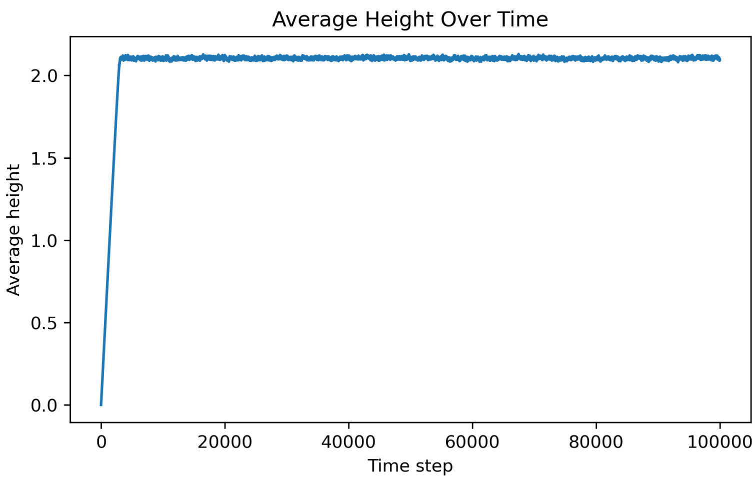
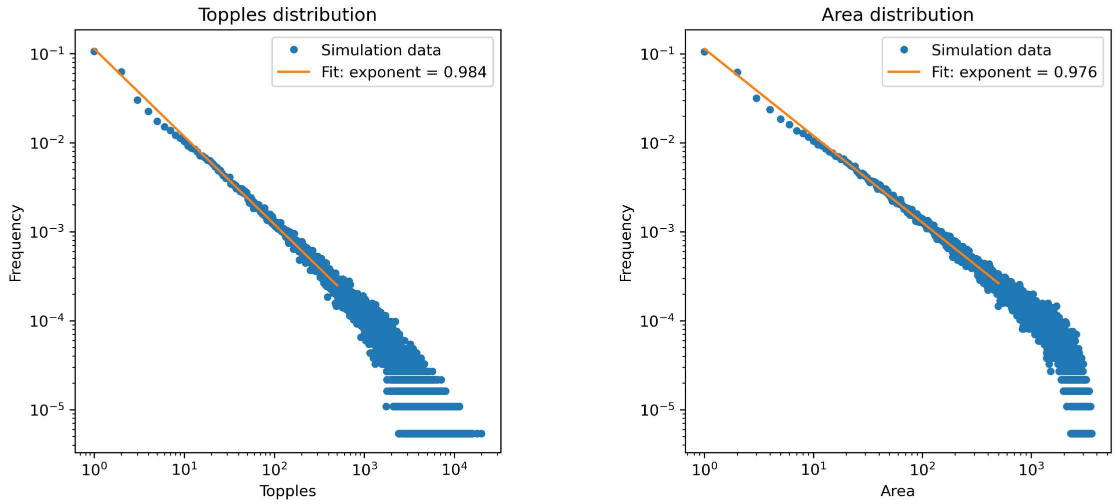


Figure 4.1: Extension: Average height over time

For avalanche statistics, the distributions of topples s, area a, loss l, and length r are expected to shift toward larger values. This is because adding three grains to the same site creates a stronger local perturbation, which may trigger larger avalanches.

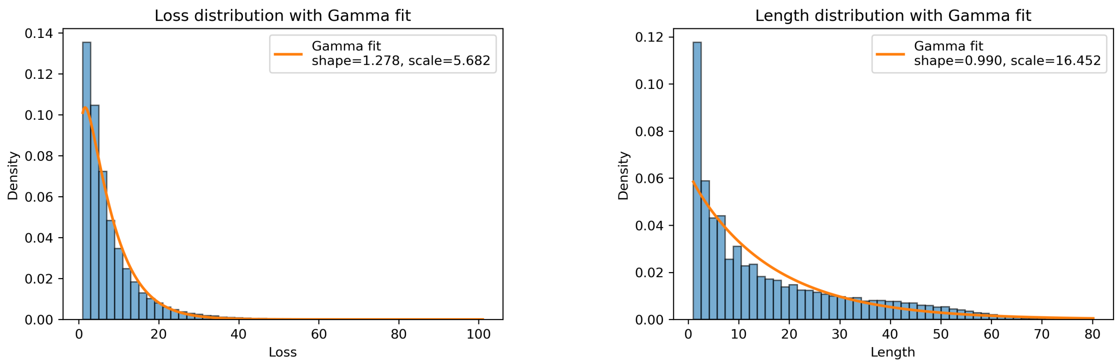
From the simulation (as shown in Figure 4.2 and Figure 4.3), the results are generally consistent with the previous prediction. First, topples and area still show approximate power-law behaviour in the extension model. The topples exponent in the extension model is approximately 0.984, which is smaller than 1.070 in the original model. The area exponent is approximately 0.976, which is also smaller than 1.059 in the original model. A smaller exponent means that the distribution decays more slowly, so large avalanches are more likely to occur in the extension model. Second, loss and length can still be fitted by Gamma distributions. In the extension model, the scale parameter of loss is 5.682, which is larger than 4.904 in the original model. The scale parameter of length is 16.452, which is also larger than 15.090 in the original model. This suggests that avalanches in the extension model are more likely to produce larger boundary loss and spread over a larger spatial range.



(a) Extension: Topples distribution

(b) Extension: Area distribution

Figure 4.2: Extension: Distribution fit for topples and area



(a) Extension: Loss distribution

(b) Extension: Length distribution

Figure 4.3: Extension: Distribution fit for topples and area

For correlations, different avalanche observables are still expected to show mainly positive correlations. In particular, topples, area, and radius should keep strong positive correlations, because a larger avalanche usually contains more topplings, affects more lattice sites, and spreads farther across the lattice.

From simulation, the correlations between topples & area and area & length are 0.921 and 0.890 respectively, which are similar with the results in the original model. Furthermore, the extension model does not strongly change the correlation structure between avalanche observables, then the results still support the previous prediction.

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